Failover in Multi-homed Topology with FRRouting and Wireshark

To validate failover within our multi-homed topology, I needed to observe how the network reacts when a primary communication link is unexpectedly severed. In a robust eBGP environment, if the connection to the primary Internet Service Provider is lost, the routing protocol must instantly detect the failure, flush the dead routes, and promote a backup path to maintain connectivity.

I simulated a physical cable cut by administratively disabling the branch router's interface that connects to ISP 1. I accessed the FRRouting command-line interface on the branch container and executed the following interface shutdown commands:

branch# configure terminal

branch(config)# interface eth1

branch(config-if)# shutdown

branch(config-if)# exit

branch(config)# exit

By shutting down eth1, I isolated the effect of a hard outage without restarting the entire router, providing a true test of BGP reconvergence.

Establishing the Traffic Steering Baseline

Before testing the failure, it was necessary to prove that the branch was actually steering traffic through the intended primary path. Because multi-homed branches must steer traffic, I had previously configured a route-map that applied a Local-Preference attribute of 200 to all incoming routes from ISP 1 (AS 100).

By checking the BGP routing table prior to the link failure, I confirmed this traffic steering was functioning perfectly.

```
bala@bala-IdeaPad-Slim-3-15IAH8:/data/bala/Projects/wipro-5g-capstone$ docker exec -it clab-wipro-5g-bgp-branch vtysh
% Can't open configuration file /etc/frr/vtysh.conf due to 'No such file or directory'.

Hello, this is FRRouting (version 8.4_git).
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branch# show ip bgp
BGP table version is 2, local router ID is 10.1.1.1, vrf id 0
Default local pref 100, local AS 65001
Status codes:  s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self
Origin codes:  i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop            Metric LocPrf Weight Path
*  8.8.8.8/32       192.168.12.1                           0 200 888 i
*>                  192.168.11.1                  200      0 100 888 i
*> 10.1.1.0/24      0.0.0.0                  0         32768 i

Displayed  2 routes and 3 total paths
branch#
```

Figure 1: Pre Failure Primary Path

As shown in the terminal capture above, the branch router possessed two paths to reach the 8.8.8.8/32 internet destination. The path via next-hop 192.168.11.1 (ISP 1) was actively selected as the absolute best path (indicated by the *> marker) precisely because its Local Preference of 200 outweighed the default preference of the ISP 2 path. This established ISP 1 as our primary communication link.

Validating High Availability and Redundancy

The core objective of this test was to ensure that if the branch-to-ISP 1 connection drops, the network does not suffer a total blackout. Immediately after executing the shutdown command on eth1, I ran the show ip bgp command on the branch router to capture the post-failure routing state.

```
bala@bala-IdeaPad-Slim-3-15IAH8:/data/bala/Projects/wipro-5g-capstone$ docker exec -it clab-wipro-5g-bgp-branch vtysh
% Can't open configuration file /etc/frr/vtysh.conf due to 'No such file or directory'.

Hello, this is FRRouting (version 8.4_git).
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branch# configure terminal
branch(config)# interface eth1
branch(config-if)#  shutdown
branch(config-if)#  exit
branch(config)# exit
branch# show ip bgp
BGP table version is 3, local router ID is 10.1.1.1, vrf id 0
Default local pref 100, local AS 65001
Status codes:  s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self
Origin codes:  i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop            Metric LocPrf Weight Path
*> 8.8.8.8/32       192.168.12.1                           0 200 888 i
*> 10.1.1.0/24      0.0.0.0                  0         32768 i

Displayed  2 routes and 2 total paths
branch#
```

Figure 2: Post Failure Backup Path

The evidence in the routing table confirms a flawless failover execution. The BGP process immediately recognized that the 192.168.11.1 next-hop was dead and removed the primary route from the table entirely. The *> best-path marker instantly shifted to the backup next-hop at 192.168.12.1 (ISP 2).

This demonstrates that the eBGP peering from the branch to two ISPs provides true high availability. The policy dictating normal operation seamlessly gave way to the surviving ISP 2 path, proving that the branch would maintain internet access during a severe ISP 1 outage.